# CAMREN KHOURY

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## EDUCATION

**Clemson University – Clemson, SC**
Bachelor of Science in Computer Engineering (In Progress)
Bachelor of Science in Electrical Engineering (In Progress)
Expected Graduation: May 8, 2026

Engineering student with hands-on experience in embedded systems, FPGA development, computer vision, and high-performance computing. Seeking embedded/IoT, firmware, automation, or Intelligent Systems roles.

## EXPERIENCE

**Clemson University – IS-WiN Lab | Clemson, SC** **January 2026 - Present**
*Undergraduate Research Assistant – FireSense Project*

- Evaluate multimodal AI outputs for wildfire detection using RGB and thermal imagery.
- Validate labels and log results to support fire perception research.
- Assist in dataset quality control for AI model assessment.

**Athena Consulting Group (ACG) | North Charleston, SC** **May 2022 – August 2022**
*Cybersecurity Intern*

- Assisted cybersecurity engineers with basic vulnerability scans and patch-compliance workflow support.
- Worked on weekly presentations, graphic design, and spreadsheets.

**North Charleston Aquatic Center | North Charleston, SC** **March 2021 – August 2024**
*Lifeguard + Swim Coach*

- Monitored pool safety and maintained operational protocols during high-traffic periods.
- CPR and AED certified for four years.
- Coached youth swimmers ages 5–16, including skill development and competitive readiness.

## TECHNICAL PROJECTS

**Automated Pool Vision & Control System – Senior Design I (ECE)**

- Built a **MATLAB** computer-vision pipeline (HSV masking, morphology, centroid extraction, homography calibration, shot geometry).
- Developed a **MATLAB** App Designer UI for calibration, real-time ball detection, and automated shot execution.
- Implemented **Arduino**-based PWM control for DC motor, servo, and solenoid firing and tuned **PID** based closed-loop control via **Simulink** HIL.
- Integrated full workflow from **computer-vision** detection → shot planning → physical execution, enabling autonomous play.

**Water-Quality Sensor System — CEDC Technical Solutions Director**

- Lead a 6-member team overseeing project schedules, deliverables, and weekly engineering meetings. Designed new organization website layout; improved clarity and navigation based on internal feedback. Initiated design research for **microcontroller-based** modular water-quality sensor system.
- Researching modular sensor platform for turbidity, chlorine, pH, and TDS **trend detection**.
- Supporting **PCB** architecture planning and early deployment feasibility analysis.

**Raspberry-Pi High-Performance Computing Cluster**

- Built and configured a **Raspberry-Pi** compute cluster including **MPI**, static-IP networking, SSH key authentication, and **NFS** shared-filesystem integration.
- Benchmarked cluster performance using **HPL** and performed workload-distribution tuning to evaluate scaling efficiency.
- Competed in the 2025 SBCC Small-Board Cluster Competition, operating an 11-node cluster and running distributed workloads including password-cracking and ParFEMWARP.

## TECHNICAL SKILLS

**Programming**: C, Python, C++, Swift, MATLAB, SQL, HTML/CSS/JS, OCaml, R
**Hardware**: Microcontrollers (Arduino/RPi/ESP32), Circuit Design, PCB Fabrication, Verilog HDL, Simulink
**Tools**: Git/GitHub, Linux, MATLAB App Designer, Excel Analytics, AI Workflow Automation
**Groups and Affiliations:** IEEE Student Branch Member • Clemson Engineers for Developing Communities • Clemson Club Swim Team • Clemson SBCC High- Performance Cluster Team